

Introduction to Self-Supervised Learning

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(Slides courtesy of Peyman Tahghighi)

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Learning objectives

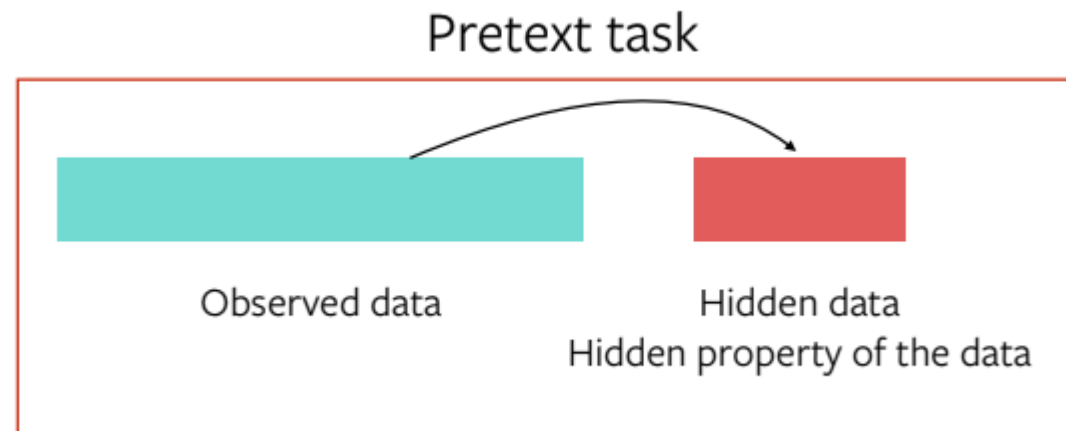
- What is self-supervised learning?
- How we can use it in NLP and Vision?
- How it can benefit us?

Self-supervised learning: The dark matter of intelligence

- Supervised learning is a bottleneck for building more intelligent generalist models that can do multiple tasks and acquire new skills without massive amounts of labelled data.
- As babies, we learn how the world works largely by observation.
- We learn new skills by short training, e.g. driving.
- A working hypothesis is that generalized knowledge about the world, or common sense, forms the bulk of biological intelligence in both humans and animals.
- Common sense helps people learn new skills without requiring massive amounts of teaching for every single task.

What is self-supervision?

- Obtain “labels” from the data itself and leverage the underlying structure in data.
- Predict unobserved or hidden parts.



Examples of data generation in NLP

GPT

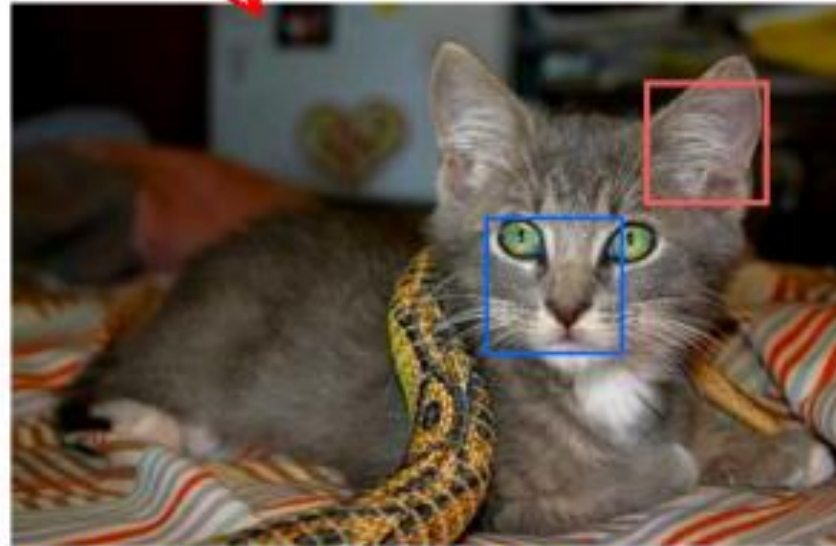
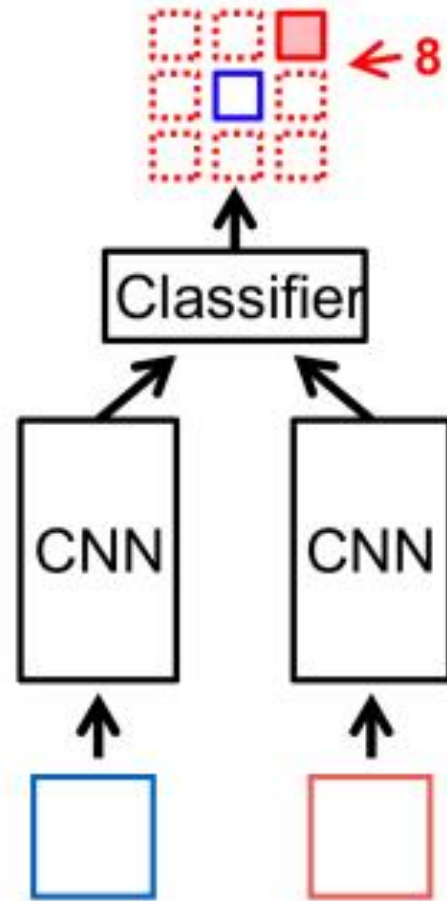
A	?							
A	quick	?						
A	quick	brown	?					
A	quick	brown	fox	?				
A	quick	brown	fox	jumped	?			
A	quick	brown	fox	jumped	over	?		
A	quick	brown	fox	jumped	over	the	?	
A	quick	brown	fox	jumped	over	the	lazy	?
A	quick	brown	fox	jumped	over	the	lazy	dog

BERT

A	?	brown	fox	?	over	the	?	dog
A	quick	?	fox	jumped	over	?	lazy	?
A	quick	brown	?	jumped	over	the	?	dog
A	?	brown	fox	?	?	the	lazy	dog

Examples of data generation in vision

Relative position of patches



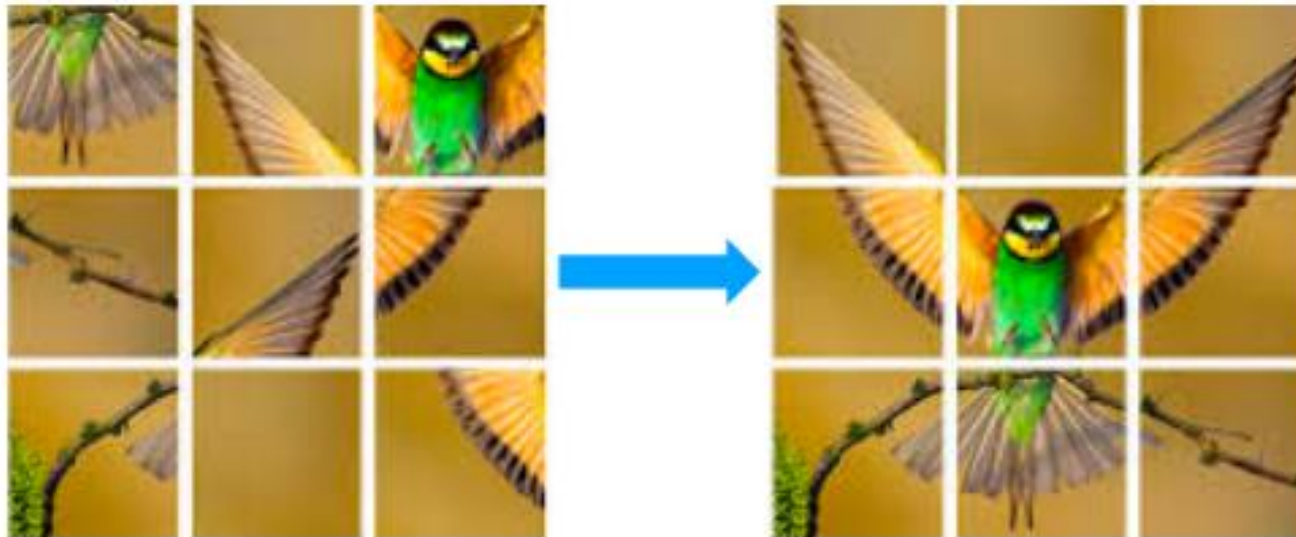
Randomly Sample Patch
Sample Second Patch

Input: Two patches

Output: 8-way classification

Examples of data generation in vision

Jigsaw puzzle



Jigsaw puzzles
(Noorozi & Favaro, 2016)

Input: nine patches
Permute using one of N
permutations

Output: N -way
classification

Set $N \ll 9!$

Examples of data generation in vision

Rotation prediction



→ 0°



→ 90°



→ 180°



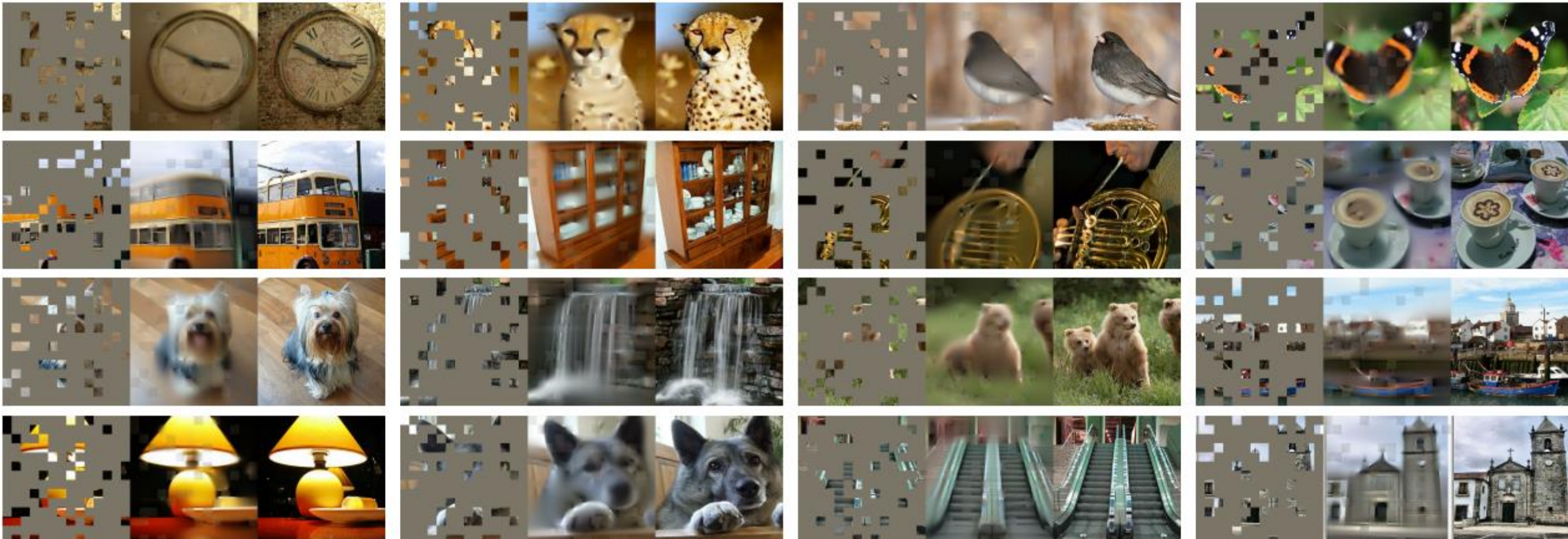
→ 270°

Input: image rotated by
[0, 90, 180, 270]

Output: 4-way classification

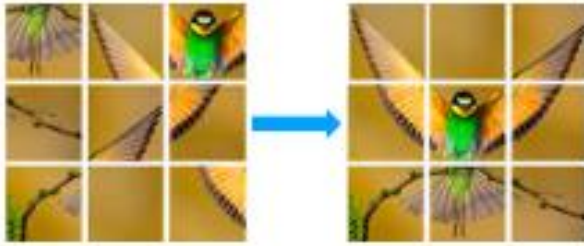
Examples of data generation in vision

Masking input



The hope of generalization

- We hope that the **pre-text** task and **downstream** task are aligned.

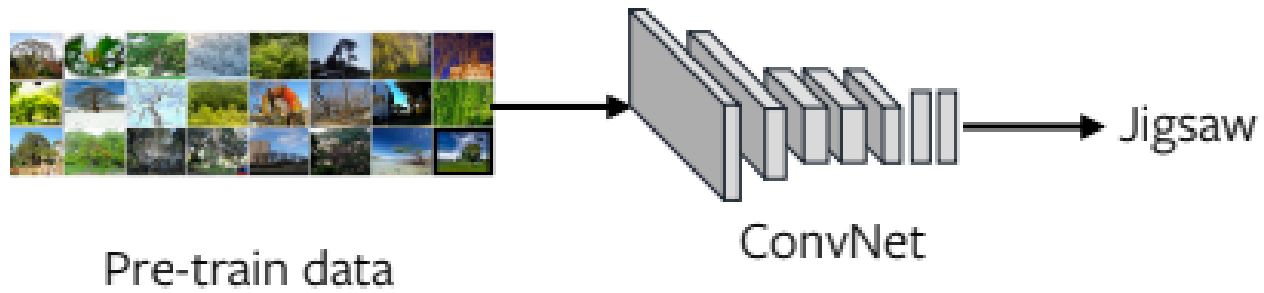


Pre-training
Self-supervised

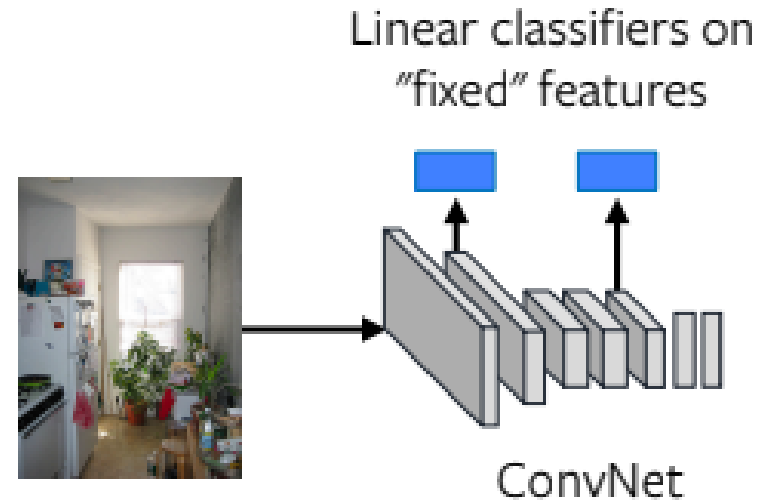


Transfer Tasks

The hope of generalization

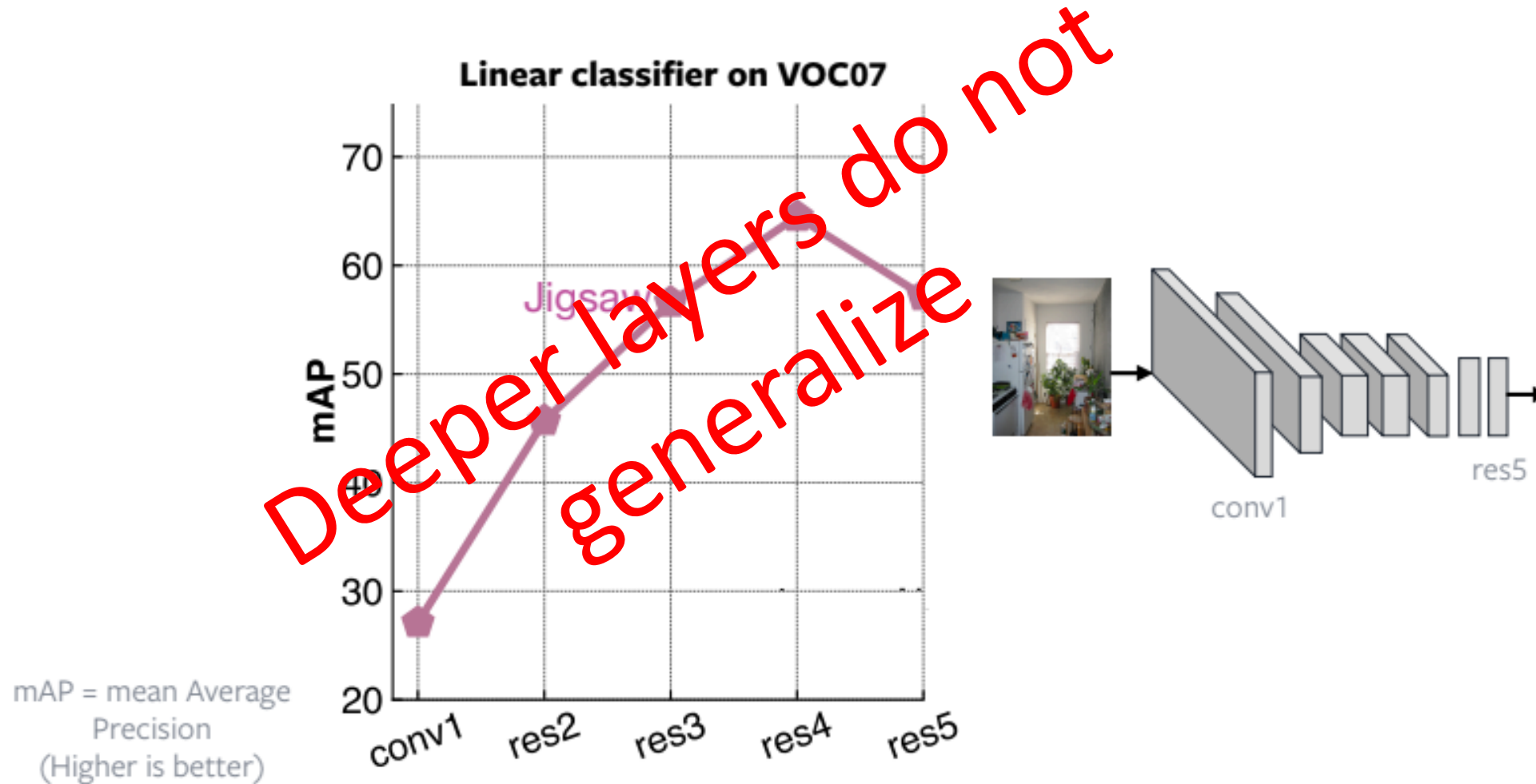


Pre-training
Weak or self-supervised



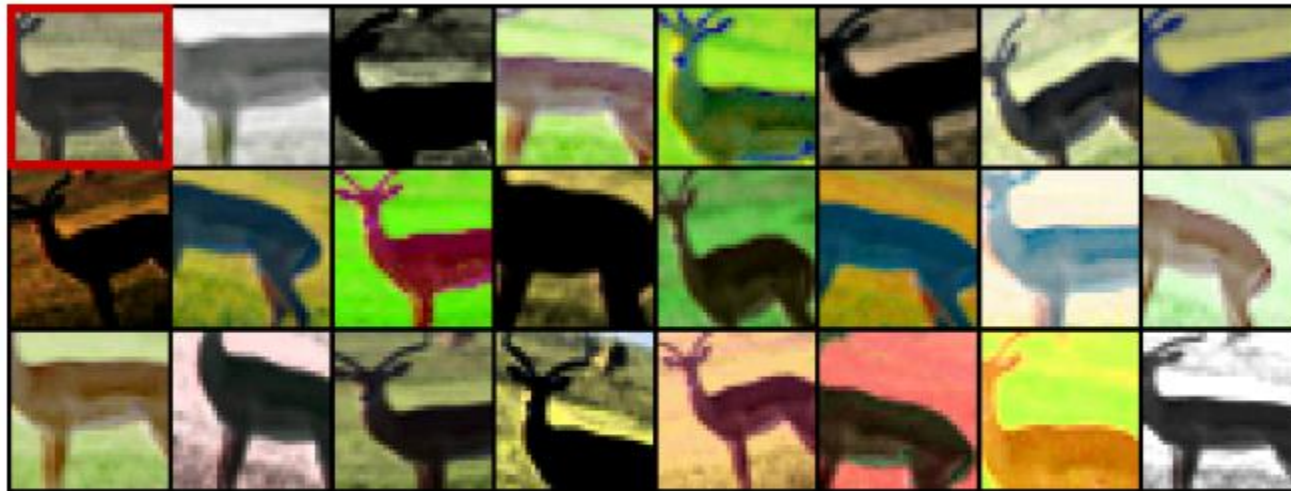
Transfer

The hope of generalization



Pretrained features should

- Represent how images relate to each other.
- Be robust to “nuisance factors”
 - E.g., exact location, lighting, texture, color,...



Learn features such that:

$$f_{\theta}(I) = f_{\theta}(\text{augment}(I))$$

Figure from Dosovitskiy et al., 2014

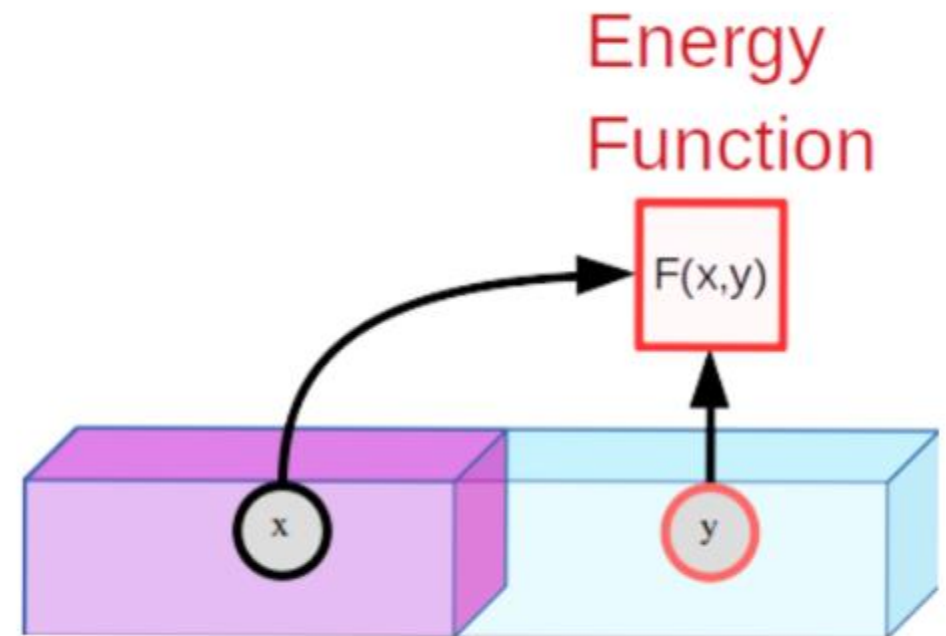
Self-supervision NLP vs Vision

	NLP	Vision
Type	Discrete	Continuous
Uncertainty		

Number of possible outcomes **can't** be enumerated in vision.

Energy-based model (EBM)

- An EBM is a trainable system that, given two inputs, x and y , tells us how incompatible they are with each other.
- Two steps of training:
 - Showing **compatible** x and y , produces low energy
 - For incompatible examples of y , give high energy

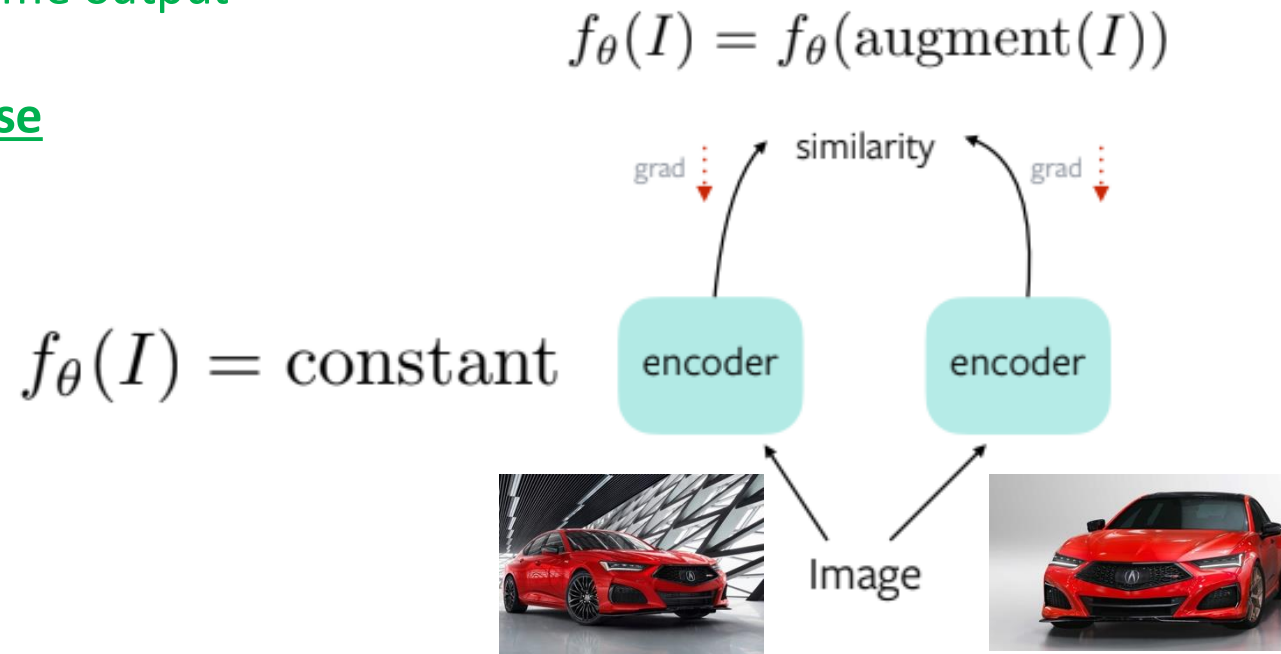


Join Embedding, Siamese networks

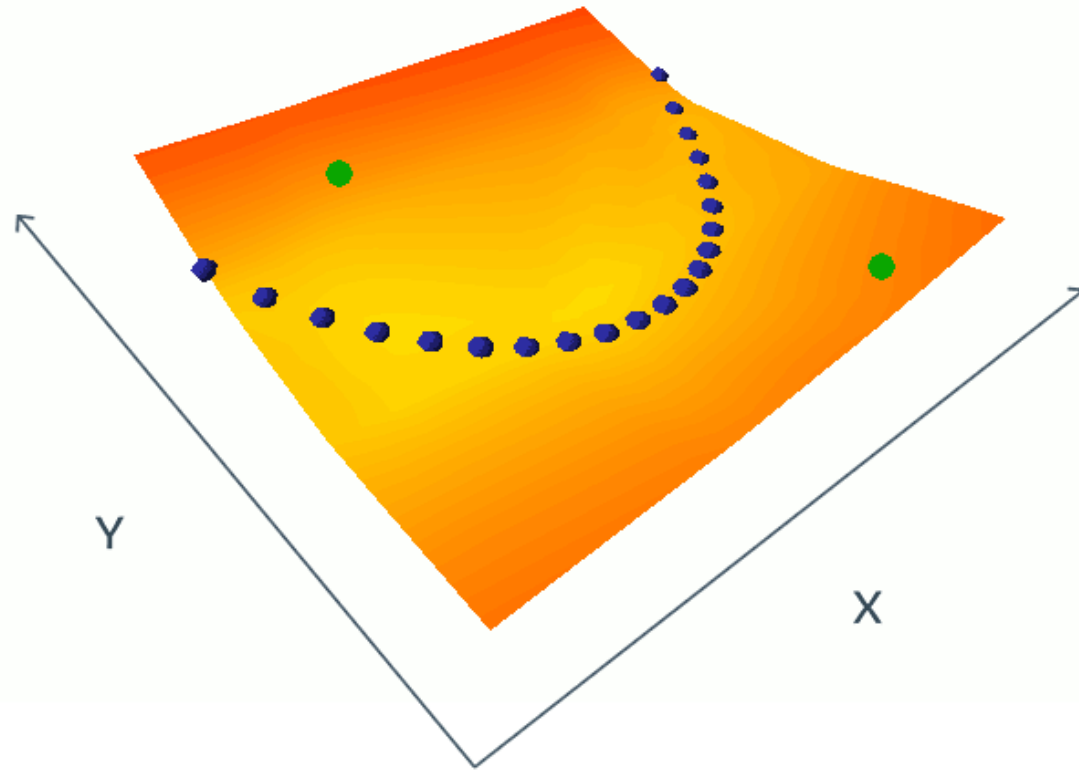
What could go wrong here?

Network ignoring input and
generating same output
regardless.

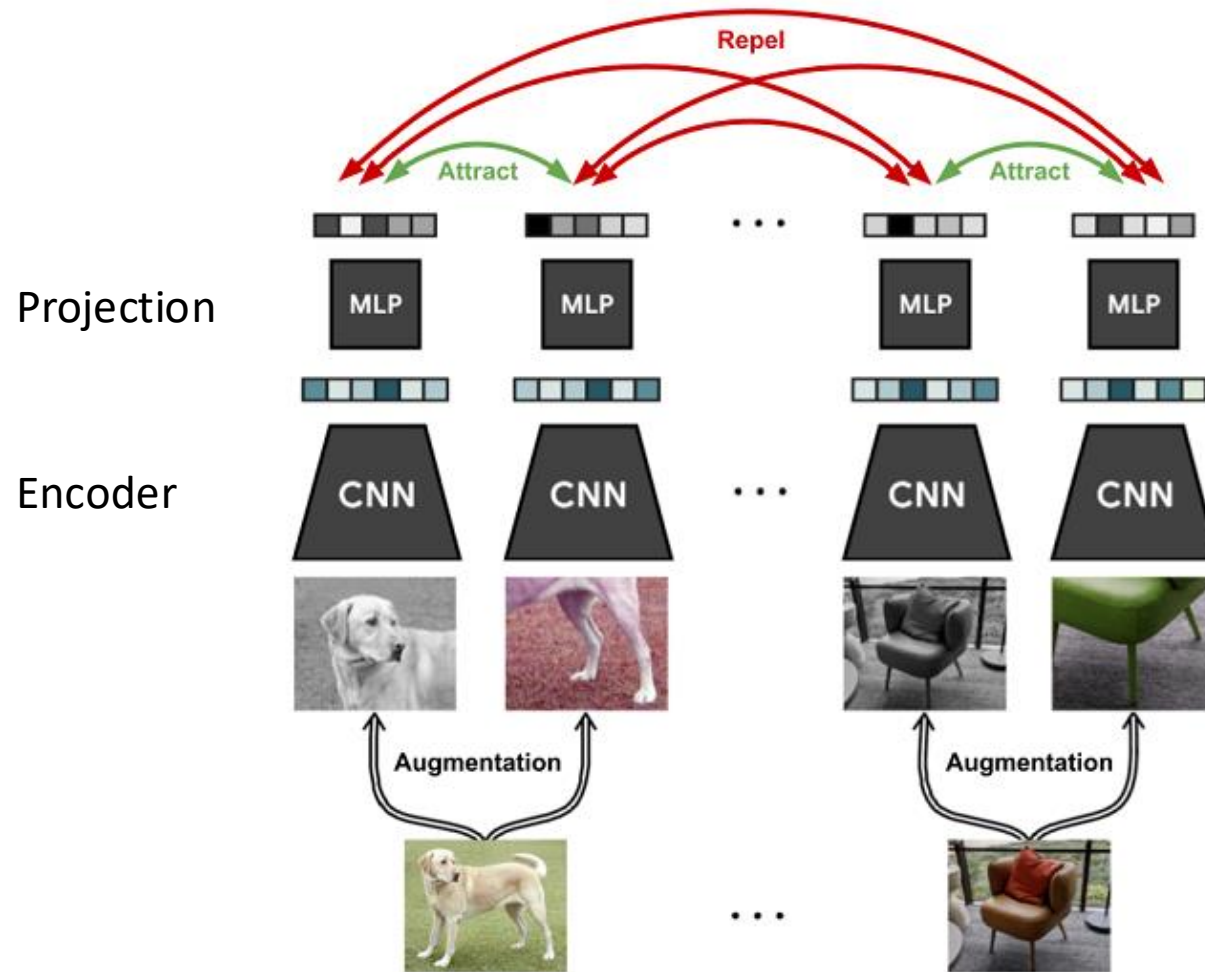
Called **Collapse**



Contrastive energy-based SSL

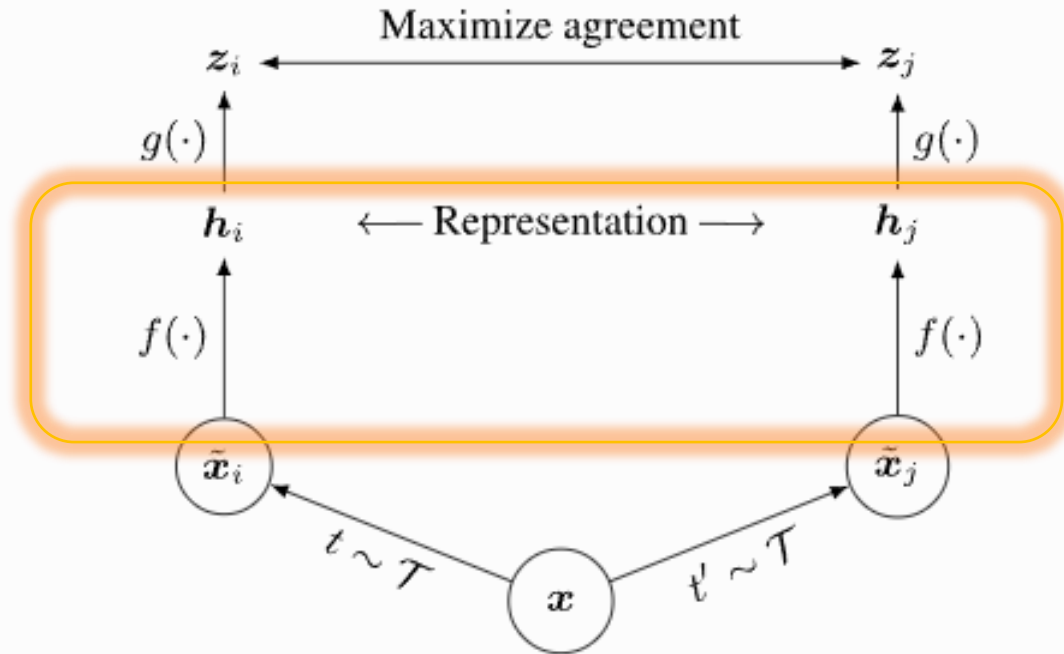


SimCLR: A Simple Framework for Contrastive Learning



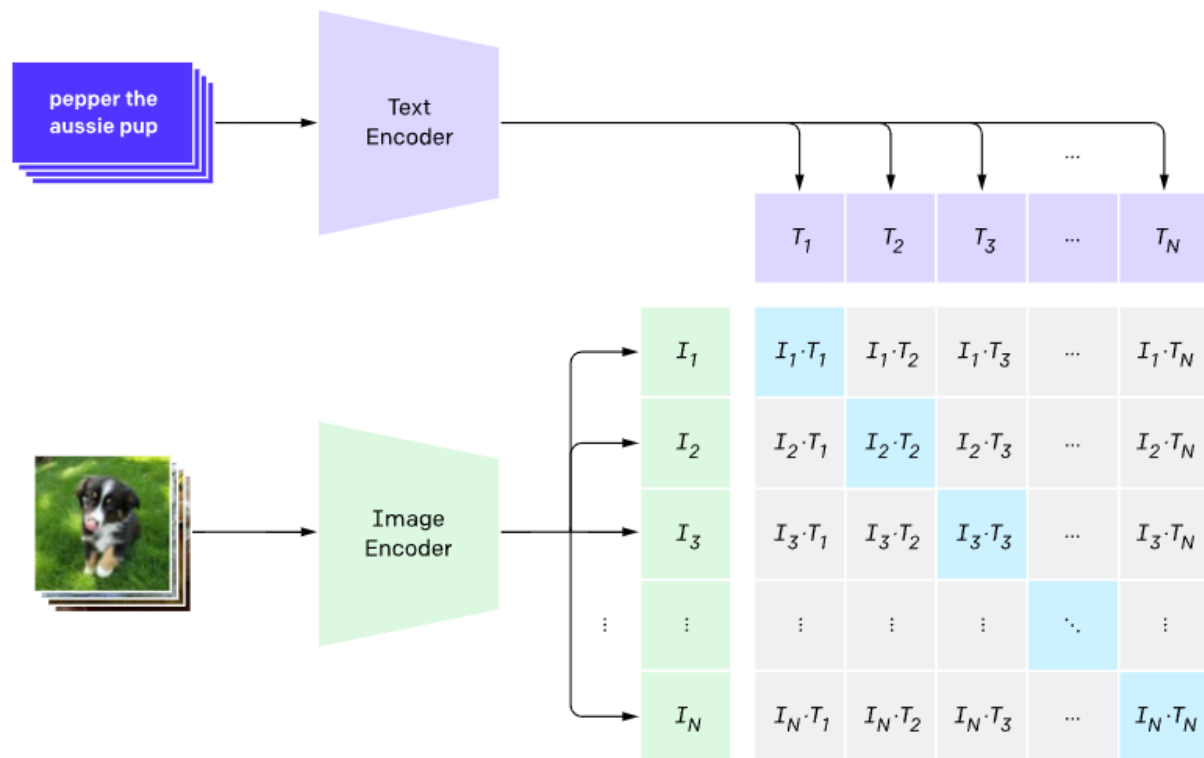
SimCLR: A Simple Framework for Contrastive Learning

Used a pre-trained weights

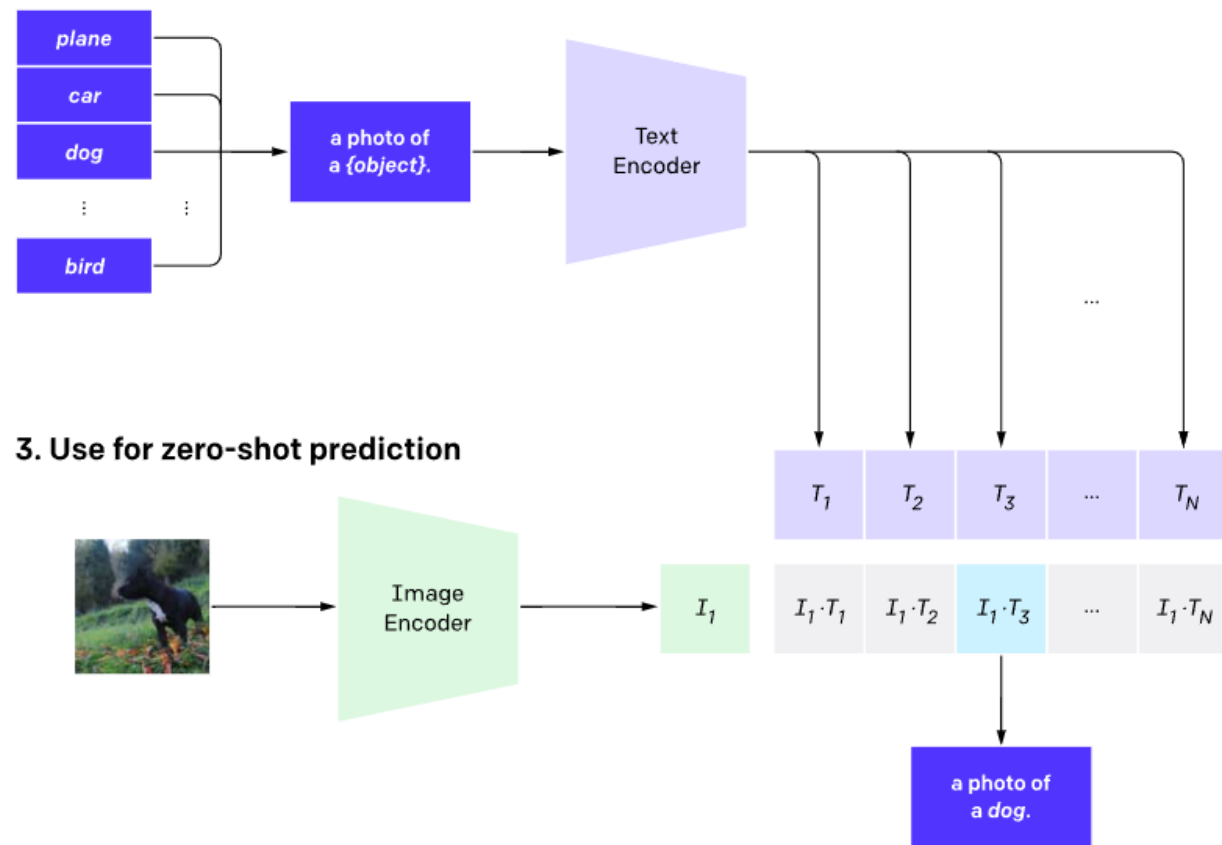


CLIP: Contrastive Language-Image Pretraining

1. Contrastive pre-training



2. Create dataset classifier from label text



Clip solves:

- Costly dataset: Can be trained using image-text pair found on internet.
- Adaptation: Easily adopted to other unseen datasets

CLIP: Contrastive Language-Image Pretraining

SUN397

television studio (90.2%) Ranked 1 out of 397 labels



✓ a photo of a **television studio**.

✗ a photo of a **podium indoor**.

✗ a photo of a **conference room**.

✗ a photo of a **lecture room**.

✗ a photo of a **control room**.

Summary

- Self-supervised learning means using the underlying structure of data to obtain data.
- Self-supervised learning is effective in both NLP and Vision
- Self-supervised learning is mostly used as pretraining phase

Useful resources

- https://www.youtube.com/watch?v=8L10w1KoOU8&t=486s&ab_channel=AlfredoCanziani
- <https://ai.meta.com/blog/self-supervised-learning-the-dark-matter-of-intelligence/>